

Unfunded Fiscal Policy with Heterogeneity in Household Portfolios*

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Abstract

When do fiscal deficits become inflationary? In a heterogeneous-agent New Keynesian model in which households hold liquid nominal government liabilities and illiquid real capital, the answer depends on fiscal backing. Deficits that are not funded by higher expected future primary surpluses reduce household demand for nominal government bonds and shift demand toward goods, so the price level rises to clear the goods market. Funded deficits generate much smaller responses because higher expected future primary surpluses sustain household demand for nominal government bonds. Quantitatively, unfunded government purchases and unfunded transfers produce nearly identical short-run responses and have much larger effects on inflation and output than funded deficits. Unfunded transfers and unfunded government purchases lower consumption inequality. Funded transfers raise low-wealth households' consumption more than funded government purchases. When households can adjust illiquid assets more often, the difference between the effects of funded and unfunded deficits on inflation and real activity becomes more pronounced.

JEL codes: E62, E31, E52, D31, G11

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Introduction

Large fiscal expansions and elevated inflation after the COVID-19 pandemic have renewed interest in a basic question: under what circumstances do fiscal deficits raise inflation?

A fiscal deficit does not, by itself, imply inflation. When higher government debt is expected to be matched by higher future primary surpluses, the fiscal expansion mainly changes the timing of fiscal adjustment rather than the valuation of government debt. Fiscal explanation of inflation requires that fiscal deficits have real effects that are not neutralized by expected future surpluses. Existing work proposes two distinct channels for fiscal deficits to affect inflation and economic activity.

One channel comes from the Fiscal Theory of the Price Level (FTPL). The real value of outstanding nominal government debt must be consistent with expected future fiscal backing. When higher deficits are not expected to be matched by higher future primary surpluses, households are no longer willing to hold nominal government bonds at the pre-shock real value. The price level rises to reduce the real value of nominal government debt until household asset demand and fiscal backing are again consistent with equilibrium (Cochrane, 2001; Leeper, 1991; Woodford, 1995). This logic has been used to interpret both earlier episodes of fiscally driven inflation and the post-COVID inflation surge (Balke and Zarazaga, 2024; Barro, 2022; Bianchi et al., 2023; Bianchi and Ilut, 2017; Hall and Sargent, 2011).

An alternative channel comes from heterogeneous-household models with nominal rigidities. Fiscal deficits become inflationary when they shift resources toward households with high marginal propensities to consume, that is, households whose current consumption responds positively to additional income. The resulting rise in aggregate demand is stronger than in representative-agent environments, and under sticky prices or wages it generates quantitatively important inflation responses even without making fiscal backing central to the analysis (Angeletos et al., 2024). This literature therefore suggests that heterogeneity alone can account for substantial inflationary effects of fiscal deficits. That conclusion is typi-

cally derived in environments in which households save in a single liquid nominal government bond and do not choose whether to reallocate wealth into illiquid real capital.

This paper asks whether redistribution alone remains sufficient to generate large inflation responses once households can also shift wealth between liquid nominal and illiquid real assets. Building on the framework in [Bayer et al. \(2024\)](#), I develop a heterogeneous-agent New Keynesian (HANK) model with long-term nominal government debt, sticky prices and wages, and endogenous household portfolio choice. Households can save or borrow liquid nominal assets, but can also hold illiquid real capital that can be adjusted only intermittently. Fiscal policy operates through government purchases and transfers, including both uniform lump-sum transfers and transfers targeted to different household groups. In addition, I compare funded deficits, which are backed by higher expected future primary surpluses, with unfunded deficits, for which future primary surpluses do not provide sufficient backing.

When households save only in liquid nominal assets, they choose only between current consumption and liquid nominal asset holdings. When households can also hold illiquid real capital, they can reallocate wealth toward illiquid real assets, which dampens the immediate effect on aggregate demand and inflation. Redistribution alone therefore do not generate large inflation responses once households can shift wealth between liquid nominal bonds and illiquid real assets. Once households can rebalance between liquid nominal assets and illiquid real assets, fiscal backing becomes quantitatively central .

The main findings are fourfold. First, unfunded deficits generate much larger aggregate responses than funded deficits. Second, when deficits are unfunded, government purchases and transfers generate very similar aggregate responses. Third, unfunded fiscal shocks lower consumption inequality by raising the consumption of households in the bottom and middle of the distribution and lowering the consumption of households at the top. Fourth, when deficits are funded, the type of fiscal expansion matters. Funded transfers provide immediate support to the consumption of households with high marginal propensities to consume (MPCs), while funded government purchases raise the consumption of households with low MPCs.

Lessening the frictions that restrict households' access to the illiquid capital market enlarges the quantitative gap between the responses to funded and unfunded deficits: unfunded deficits generate larger responses, while funded deficits generate smaller ones. This result suggests that household heterogeneity alone does not replace the FTPL explanation of inflation. Once households can reallocate wealth across liquid nominal bonds and illiquid real assets, fiscal backing remains quantitatively central for inflation. A decline in asset-holding frictions has two effects. First, households can self-insure idiosyncratic risk more effectively and smooth consumption more over time. This dampens responses to funded deficits, because households have less need to front-load consumption out of additional disposable resources. Second, households can rebalance wealth away from nominal government bonds more easily when fiscal backing weakens. For unfunded deficits, the decline in the real value of nominal government bonds creates a stronger incentive to shift wealth toward illiquid real assets, raising investment and thereby increasing output and inflation.

The paper contributes to two strands of literature. First, it links the funded-versus-unfunded deficit distinction in the FTPL literature to the heterogeneous-household literature on fiscal transmission and inflation. The FTPL literature shows that the real value of nominal government debt depends on expected fiscal backing, often in representative-agent settings or policy-regime analyses (Bianchi and Ilut, 2017; Cochrane, 2001; Hall and Sargent, 2011; Leeper, 1991; Woodford, 1995). Closest in spirit, Leeper et al. (2017) study fiscal financing and monetary-fiscal regimes in an estimated DSGE model with savers and non-savers. Their focus is on government spending multipliers across regimes. This paper takes the fiscal-backing insight to a heterogeneous-agent economy with a full distribution of household portfolios, in which households can rebalance between liquid nominal government liabilities and illiquid real capital. In this environment, fiscal backing affects aggregate demand through household portfolio choices and also shapes the distribution of consumption. This mechanism differs from existing heterogeneous-household work, which explains inflation mainly through redistribution across households with different marginal propensi-

ties to consume ([Angeletos et al., 2024](#); [Kaplan, 2025](#)). I show that once households can reallocate wealth between liquid nominal assets and illiquid real assets, redistribution alone does not fully account for the inflationary effects of fiscal deficits. Fiscal backing remains quantitatively central.

Second, the paper contributes to the literature on household portfolios and policy transmission. Quantitative work with liquid and illiquid assets shows that household portfolio structure matters for aggregate dynamics and policy transmission ([Bayer et al., 2023, 2024](#); [Luetticke, 2021](#)). Building on that insight, this paper provides a quantitative framework in which portfolio reallocation and fiscal backing jointly shape the transmission of fiscal deficits.

The remainder of the paper is organized as follows. Section 1 presents the model. Section 2 discusses the calibration. Section 3 reports the quantitative results. Section 4 presents sensitivity checks. Section 5 reports robustness checks. Section 6 concludes.

1 Model

I build a heterogeneous-agent New Keynesian model with incomplete markets, long-term nominal government debt, and endogenous household portfolio choice. Households save in two assets: a liquid nominal asset and illiquid real capital. The government runs deficits through government purchases and transfers and distinguishes between funded deficits, for which expected future primary surpluses provide sufficient backing, and unfunded deficits, for which expected future primary surpluses do not offset current deficits. I consider transfers may be distributed uniformly across households or are targeted to specific household groups. Firms operate under monopolistic competition with sticky prices and wages. Banks intermediate household liquid assets into long-term nominal government debt.

The household side builds on [Bayer et al. \(2024\)](#), the banking and long-term debt side follows [Bayer et al. \(2023\)](#), and the funded-versus-unfunded fiscal structure follows [Balke and Zarazaga \(2024\)](#). Appendix A provides additional details on preferences, taxation, firm-side

optimization, and wage setting.

1.1 Government and Fiscal Backing

The government issues long-term nominal debt, levies taxes on labor and profit income, and runs deficits through government purchases and transfers. I distinguish between funded government purchases and transfers, for which expected future primary surpluses provide sufficient backing, and unfunded government purchases and transfers, for which expected future primary surpluses do not provide sufficient backing. Monetary policy follows a passive Taylor rule.

1.1.1 Fiscal Authority

Let $Bgov_{t+1}$ denote total government debt at the end of period t . I decompose total government debt into funded debt, $BFgov_{t+1}$, and unfunded debt, $BUGov_{t+1}$:

$$Bgov_{t+1} = BFgov_{t+1} + BUGov_{t+1}. \quad (1)$$

Total government debt evolves according to

$$Bgov_{t+1} = G_t + Transfer_t - Tax_t + \frac{R_t^b}{\pi_t} Bgov_t, \quad (2)$$

where G_t is total government consumption, $Transfer_t$ is total government transfers, Tax_t is tax revenue, R_t^b is the gross nominal interest rate on government debt, and π_t is gross inflation.

Funded debt evolves according to

$$BFgov_{t+1} = GF_t + TF_t - \chi_1 Tax_t + \frac{R_t^b}{\pi_t} BFgov_t, \quad (3)$$

where GF_t and TF_t denote the funded components of government purchases and transfers,

and $\chi_1 \in [0, 1]$ is the share of tax revenue allocated to the funded debt component. I set $\chi_1 = BF\bar{gov}/B\bar{gov}$ in steady state. The unfunded component is then

$$BUgov_{t+1} = Bgov_{t+1} - BFgov_{t+1}. \quad (4)$$

Government purchases and transfers are split into funded and unfunded components:

$$G_t = GF_t + GU_t, \quad (5)$$

$$Transfer_t = TF_t + TU_t. \quad (6)$$

Out of steady state, the four components evolve according to

$$\frac{GU_t}{\bar{GU}} = \left(\frac{GU_{t-1}}{\bar{GU}} \right)^{\rho_{GU}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{GU})\gamma_{YGU}} \left(\frac{BUgov_t}{B\bar{Ugov}} \right)^{(1-\rho_{GU})\gamma_{BGU}} \epsilon_t^{GU}, \quad (7)$$

$$\frac{GF_t}{\bar{GF}} = \left(\frac{GF_{t-1}}{\bar{GF}} \right)^{\rho_{GF}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{GF})\gamma_{YGF}} \left(\frac{BFgov_t}{B\bar{Fgov}} \right)^{(1-\rho_{GF})\gamma_{BGF}} \epsilon_t^{GF}, \quad (8)$$

$$\frac{TU_t}{\bar{TU}} = \left(\frac{TU_{t-1}}{\bar{TU}} \right)^{\rho_{TU}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{TU})\gamma_{YTU}} \left(\frac{BUgov_t}{B\bar{Ugov}} \right)^{(1-\rho_{TU})\gamma_{BTU}} \epsilon_t^{TU}, \quad (9)$$

$$\frac{TF_t}{\bar{TF}} = \left(\frac{TF_{t-1}}{\bar{TF}} \right)^{\rho_{TF}} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_{TF})\gamma_{YTF}} \left(\frac{BFgov_t}{B\bar{Fgov}} \right)^{(1-\rho_{TF})\gamma_{BTF}} \epsilon_t^{TF}. \quad (10)$$

The parameters γ_{BG_-} and γ_{BT_-} measure how strongly government purchases and transfers respond to debt. A higher value implies that higher debt is followed by a larger increase in the corresponding spending component. This makes it harder for future primary surpluses to rise enough to provide sufficient backing for the debt. Lower values therefore correspond to sufficient fiscal backing, whereas higher values correspond to insufficient fiscal backing. In the calibration, the funded coefficients, γ_{BGF} and γ_{BTF} , are set low enough that fiscal policy remains passive, whereas the unfunded coefficients, γ_{BGU} and γ_{BTU} , are set high enough that fiscal policy is active.

Taxation. The tax system follows [Bayer et al. \(2024\)](#). The average tax rate τ_t evolves according to

$$\frac{\tau_t}{\bar{\tau}} = \left(\frac{\tau_{t-1}}{\bar{\tau}} \right)^{\rho_\tau} \left(\frac{BFgov_t}{BFgov} \right)^{(1-\rho_\tau)\gamma_B^\tau} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_\tau)\gamma_Y^\tau}, \quad (11)$$

and tax progressivity follows

$$\frac{\tau_t^P}{\bar{\tau}^P} = \left(\frac{\tau_{t-1}^P}{\bar{\tau}^P} \right)^{\rho_P} \epsilon_t^P. \quad (12)$$

Aggregate tax revenue is denoted by Tax_t . Tax revenue is raised from labor income and profits and feeds back on funded debt. Appendix [A.1](#) provides the household-level tax structure.

1.1.2 Monetary Authority

The monetary authority sets the nominal interest rate according to a Taylor-type rule,

$$\frac{R_{t+1}^b}{\bar{R}^b} = \left(\frac{R_t^b}{\bar{R}^b} \right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}} \right)^{(1-\rho_R)\theta_\pi} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_R)\theta_Y} \epsilon_t^R. \quad (13)$$

I focus on a passive monetary regime in the sense of [Leeper \(1991\)](#), $\theta_\pi < 1$.

1.2 Households and Portfolio Choice

A continuum of households faces idiosyncratic labor-income risk and chooses how much to save in a liquid nominal asset and in illiquid real capital. Workers earn labor income and receive union profits. Entrepreneurs do not work and receive firm profits.

1.2.1 Income, Assets, and Budget Constraint

The household side follows [Bayer et al. \(2024\)](#). Workers face idiosyncratic productivity risk. Entrepreneurs do not supply labor and receive non-union firm profits. Let b_{it} denote household i 's liquid asset holdings and k_{it} denote illiquid real capital. Households can borrow through the liquid asset up to the limit $\underline{B}$, while illiquid capital must remain nonnegative.

Household i 's period- t budget constraint is

$$c_{it} + b_{it+1} + q_t k_{it+1} = \frac{R_{it}}{\pi_t} b_{it} + (q_t + r_t) k_{it} + y_{it} \\ + \mathbb{I}_{h_{it} \neq 0} (1 - \tau_t) \Pi_t^U + \mathbb{I}_{h_{it} = 0} (1 - \tau_t^L) (\Pi_t^E)^{1 - \bar{\tau}^P} + Transfers_{it}, \quad (14)$$

subject to

$$k_{it+1} \geq 0, \quad b_{it+1} \geq \underline{B}. \quad (15)$$

Here, q_t is the price of capital, r_t is the dividend on capital, y_{it} is after-tax labor income, Π_t^U denotes union profits received by workers, Π_t^E denotes entrepreneurial profits, and $Transfers_{it}$ denotes government transfers.

Households choose b_{it+1} rather than direct holdings of long-term government debt. The return on the liquid asset is

$$R_{it} = \begin{cases} A_t \frac{R_t^b B_t + \pi_t [(1 - \iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F]}{B_t + q_{t-1}^\Pi} & \text{if } b_{it} \geq 0, \\ A_t \frac{R_t^b B_t + \pi_t [(1 - \iota^\Pi) q_t^\Pi + \omega^\Pi \Pi_t^F]}{B_t + q_{t-1}^\Pi} + \bar{R} & \text{if } b_{it} < 0. \end{cases} \quad (16)$$

The liquid asset return equals the average return on the aggregate liquid portfolio, composed of government debt and tradable profit-claim paper, plus a borrowing wedge for households with negative liquid asset holdings.

1.2.2 Portfolio Adjustment Friction

Only a fraction λ of households can adjust illiquid capital in a given period. Households that receive the adjustment opportunity choose both liquid assets and illiquid capital. Households that do not receive the adjustment opportunity keep illiquid capital fixed and choose only liquid asset holdings. The parameter λ governs how easily households reallocate wealth between liquid nominal assets and illiquid real capital.

1.2.3 Household Problem

Let $\Theta_t(b, k, h)$ denote the joint distribution of liquid asset holdings, capital, and idiosyncratic productivity. Define $V_t^a(b, k, h)$ as the value function for households that can adjust both assets, and $V_t^n(b, k, h)$ as the value function for households that cannot adjust capital:

$$V_t^a(b, k, h) = \max_{b'_a, k'} u[x(b, b'_a, k, k', h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_a, k', h'), \quad (17)$$

$$V_t^n(b, k, h) = \max_{b'_n} u[x(b, b'_n, k, k, h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_n, k, h'), \quad (18)$$

$$\mathbb{W}_{t+1}(b', k', h') = \lambda V_{t+1}^a(b', k', h') + (1 - \lambda) V_{t+1}^n(b', k', h'). \quad (19)$$

The cross-sectional distribution evolves according to household policy rules and the transition matrix for idiosyncratic productivity:

$$\begin{aligned} \Theta_{t+1}(b', k', h') &= \lambda \int_{b'=b_{a,t}^*(b,k,h), k'=k_t^*(b,k,h)} \Phi(h, h') d\Theta_t(b, k, h) \\ &\quad + (1 - \lambda) \int_{b'=b_{n,t}^*(b,k,h), k'=k} \Phi(h, h') d\Theta_t(b, k, h). \end{aligned} \quad (20)$$

Appendix [A.1](#) provides additional details on preferences, labor supply, tax progressivity, and income construction.

1.3 Firms, Banks, and Long-Term Government Debt

1.3.1 Firms

The production side follows a standard New Keynesian structure with monopolistic competition, sticky prices, sticky wages, and investment adjustment costs. Intermediate firms produce using labor and capital,

$$Y_t = Z_t N_t^{1-\alpha} (u_t K_t)^\alpha, \quad (21)$$

where Z_t is total factor productivity, N_t is labor input, K_t is the capital stock, and u_t is capital utilization.

Price setting and wage setting imply a standard price Phillips curve and wage Phillips curve. Capital accumulation is subject to convex adjustment costs. Appendix A.4 provides the firm-side details.

1.3.2 Banks and Long-Term Nominal Debt

Banks intermediate aggregate household liquid asset holdings into long-term nominal government debt, following Bayer et al. (2023). Government bonds have geometric maturity. Each bond pays one unit of nominal coupon per period, and a fraction δ_B matures each period. Let q_t^B denote the nominal bond price. Perfect competition in banking implies

$$R_{t+1}^b q_t^B = \mathbb{E}_t [(1 - \delta_B) q_{t+1}^B + 1]. \quad (22)$$

Real bank profits are

$$\Pi_t^B = \frac{Bgov_t}{\pi_t} \left[(1 - \delta_B) \frac{q_t^B}{q_{t-1}^B} + \frac{1}{q_{t-1}^B} - R_t^b \right]. \quad (23)$$

Banks map household liquid asset demand into the pricing of long-term nominal government debt.

1.4 Market Clearing and Equilibrium

Liquid asset market. The liquid asset market clears when

$$B_{t+1} = \mathbb{E}_t [\lambda b_{a,t}^* + (1 - \lambda) b_{n,t}^*]. \quad (24)$$

Aggregate liquid asset supply consists of long-term government debt and tradable profit claims. With the timing convention in the model, household liquid asset holdings carried

into period t equal the government debt carried into period t plus the market value of tradable profit claims purchased in period $t - 1$:

$$B_t = Bgov_t + q_{t-1}^\Pi, \quad (25)$$

where q_{t-1}^Π is the price of tradable profit claims at the end of period $t - 1$. Since the supply of tradable profit claims is normalized to one, q_{t-1}^Π is also their aggregate market value.

Tradable profit claims. The price of tradable profit claims satisfies

$$q_t^\Pi R_t^b = \mathbb{E}_t \pi_{t+1} \left[(1 - \iota^\Pi) q_{t+1}^\Pi + \omega^\Pi \Pi_{t+1}^F \right]. \quad (26)$$

Capital market. The capital market clears when aggregate household demand for illiquid capital equals the aggregate capital stock:

$$K_{t+1} = \mathbb{E}_t [\lambda k_t^* + (1 - \lambda)k]. \quad (27)$$

Goods and labor markets. The labor market clears at the equilibrium real wage. The goods market clears through the resource constraint. Market clearing in liquid assets, capital, and goods jointly determines how changes in fiscal backing translate into inflation.

A sequential equilibrium consists of household policy functions, a law of motion for the cross-sectional distribution, sequences of prices and aggregates, and government policies such that households optimize, firms and banks optimize, government budget constraints and policy rules hold, and all markets clear.

2 Calibration

I calibrate the model at quarterly frequency. The steady state is chosen to match the same broad long-run objects as [Bayer et al. \(2024\)](#), with particular attention to household

portfolio structure, the prevalence of borrowers, long-term nominal debt, and the fiscal side. The targeted moments include earnings persistence and dispersion, worker–entrepreneur transitions, the liquid–illiquid composition of household wealth, factor shares, government size, and tax progressivity.

2.1 Steady-State Calibration

The household side is disciplined by a small set of preference, income-process, and portfolio parameters. I set

$$(\xi, \gamma, \beta, \gamma_{\text{scale}}) = (4, 2, 0.98255, 0.2),$$

which pin down precautionary saving, labor supply, the capital–output ratio, and steady-state hours. The idiosyncratic income process is governed by $(\rho_h, \sigma_h) = (0.98, 0.12)$, and worker–entrepreneur transitions are set to $\iota = 1/16$ and $\zeta = 1/4500$, so that the model matches earnings persistence, earnings dispersion, and a thin entrepreneurial tail.

The portfolio structure is disciplined by three key parameters. The probability of illiquid asset adjustment is $\lambda = 0.03$, which controls how often households can rebalance between liquid nominal assets and illiquid capital. Long-term government debt has geometric maturity with $\delta_B = 1/5$, which implies an average duration of five quarters. The relative supply of liquid assets is governed by $\psi = 0.10$, and tradable profit claims are set by $(\omega_\Pi, \iota_\Pi) = (0.15, 0.016)$. Together, these parameters pin down the liquid–illiquid composition of household portfolios and the prevalence of borrowers.

On the production side, I set the capital share to $\alpha = 0.318$, consistent with a labor share of about 0.68, and quarterly depreciation to $\delta_0 = 0.0215$. Gross price and wage markups are set to $\mu^Y = \mu^W = 1.1$. On the fiscal side, tax parameters are $(\tau^L, \tau^P) = (0.645, 0.1022)$, which imply a government consumption share near 20 percent and a realistic degree of tax progressivity. I normalize steady-state gross inflation and the gross policy rate to $\pi = R^b = 1$. The borrower wedge implied by the code is $\bar{R} = \pi(1.20)^{1/4} - 1 \approx 0.0466$.

Table 1: Key calibrated parameters

Category	Symbol	Value	Role / target
Preferences	ξ	4.0	Precautionary saving
	γ	2.0	Labor supply elasticity
	β	0.98255	K/Y
Income risk	ρ_h	0.98	Earnings persistence
	σ_h	0.12	Earnings dispersion
	ι, ζ	1/16, 1/4500	Worker–entrepreneur flows
Portfolio structure	λ	0.03	Illiquid asset adjustment frequency
	δ_B	0.20	Debt maturity
	ψ	0.10	Liquid asset supply
	ω_Π, ι_Π	0.15, 0.016	Tradable profit-claim supply
Technology	α	0.318	Labor share
	δ_0	0.0215	Depreciation
	μ^Y, μ^W	1.1, 1.1	Price and wage markups
Fiscal side	τ^L, τ^P	0.645, 0.1022	Government size and progressivity
	$\bar{R}$	0.0466	Borrowing wedge

Table 1 summarizes the main calibrated parameters used in the quantitative exercises. Appendix Table 3 reports the full steady-state calibration, targets, and data sources.

2.2 Policy Rules for Funded and Unfunded Deficits

Funded and unfunded deficits are implemented through different debt-feedback coefficients in the fiscal rules. Since the rules are written for government purchases and transfers, a negative debt-feedback coefficient means that higher debt reduces purchases or transfers. This raises primary surpluses and therefore funds the initial deficit. A positive debt-feedback coefficient means that higher debt is followed by higher purchases or transfers, so the initial deficit is not backed by primary surpluses.

In the baseline calibration, funded government purchases and transfers load on negative debt-feedback coefficients, $\gamma_{BGF} = \gamma_{BTF} = -0.50$, while unfunded government purchases and transfers load on positive debt-feedback coefficients, $\gamma_{BGU} = \gamma_{BTU} = 0.50$.^{*} The other fiscal-rule coefficients are held fixed across the four deficit experiments: all government-

^{*}The results are robust to setting $\gamma_{BGU} = \gamma_{BTU} = 0$.

Table 2: Baseline policy-rule parameters

Rule	Parameters	Values
Funded government purchases	$(\rho_{GF}, \gamma_{YGF}, \gamma_{BGF})$	$(0.90, 0.50, -0.50)$
Unfunded government purchases	$(\rho_{GU}, \gamma_{YGU}, \gamma_{BGU})$	$(0.90, 0.50, 0.50)$
Funded transfers	$(\rho_{TF}, \gamma_{YTF}, \gamma_{BTF})$	$(0.90, 0.50, -0.50)$
Unfunded transfers	$(\rho_{TU}, \gamma_{YTU}, \gamma_{BTU})$	$(0.90, 0.50, 0.50)$
Monetary policy	$(\rho_R, \theta_\pi, \theta_Y)$	$(0.70, 0.50, 0.125)$

purchase and transfer rules have persistence $\rho = 0.90$ and output response $\gamma_Y = 0.50$. Thus, the main difference between funded and unfunded experiments comes from the debt-feedback coefficient. Table 2 reports the baseline fiscal-rule coefficients, the monetary policy rule, and the shock processes used in the quantitative exercises.

All exogenous shocks follow AR(1) processes. For the benchmark impulse responses, I set the innovation standard deviations to a common small value and scale the reported responses to a common shock size. Appendix Table 5 reports the full set of policy-rule and shock-process parameters.

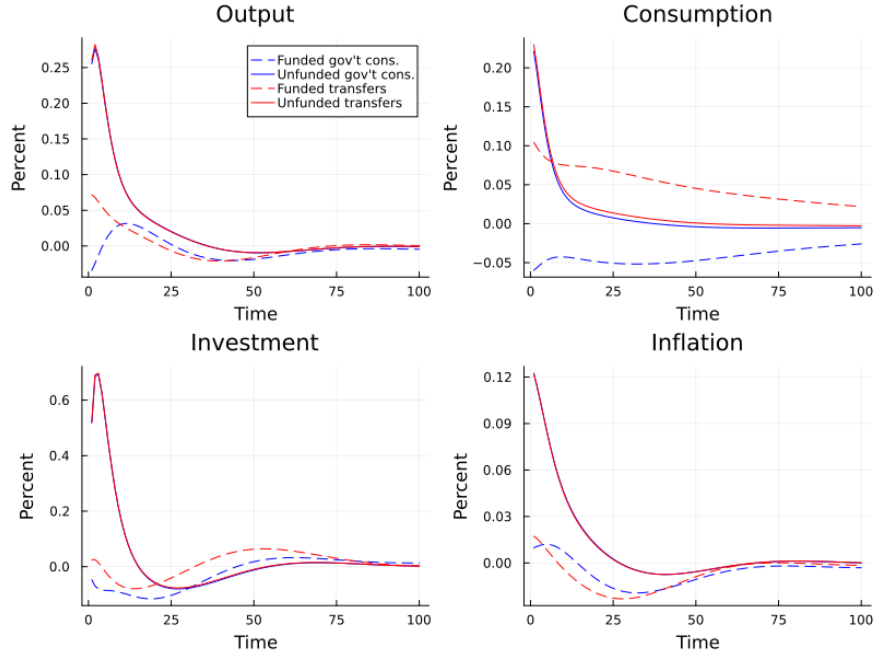
2.3 Solution Method

I solve the model following the Reiter-style approach in Bayer et al. (2024). I first compute the stationary equilibrium by iterating on household policy functions and the invariant cross-sectional distribution. I then reduce the high-dimensional distributional state, linearize the equilibrium system around the stationary equilibrium, and solve the resulting linear rational-expectations system to obtain impulse responses. Appendix C reports the numerical algorithm, state reduction procedure, and diagnostic checks.

3 Results

In this section, I examine the responses of aggregate variables to funded and unfunded fiscal shocks to transfers and government purchases. I also examine the distributional consequences

Figure 1: Impulse Response Functions (IRFs) to fiscal deficit shocks



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation, showing the effects of a 0.1-percent temporary increase in fiscal expenditures. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Red lines indicate transfers, and blue lines indicate government consumption. Transfer shocks are distributed evenly across households.

of these shocks.

3.1 Aggregate Effects on Output, Consumption, Investment, and Inflation

This section reports impulse response functions for four fiscal deficit shocks. The shocks differ along two dimensions: whether the deficit comes from government purchases or transfers, and whether the deficit is funded or unfunded.

Figure 1 reports the impulse responses of output, consumption, investment, and inflation. Two results stand out. First, unfunded fiscal deficit shocks generate much larger responses in all four variables than funded fiscal deficit shocks. Second, among unfunded shocks,

government purchases and transfers generate very similar responses.

The contrast between funded and unfunded shocks comes from fiscal backing. When the deficit is funded, higher expected future primary surpluses sustain household demand for nominal government bonds. The shift away from nominal bonds is therefore limited, goods demand rises by less, and output, consumption, investment, and inflation move only modestly. When the deficit is unfunded, weak backing reduces household demand for nominal government bonds and shifts demand toward goods. As a result, real activity rises more strongly and the price level must increase by more.

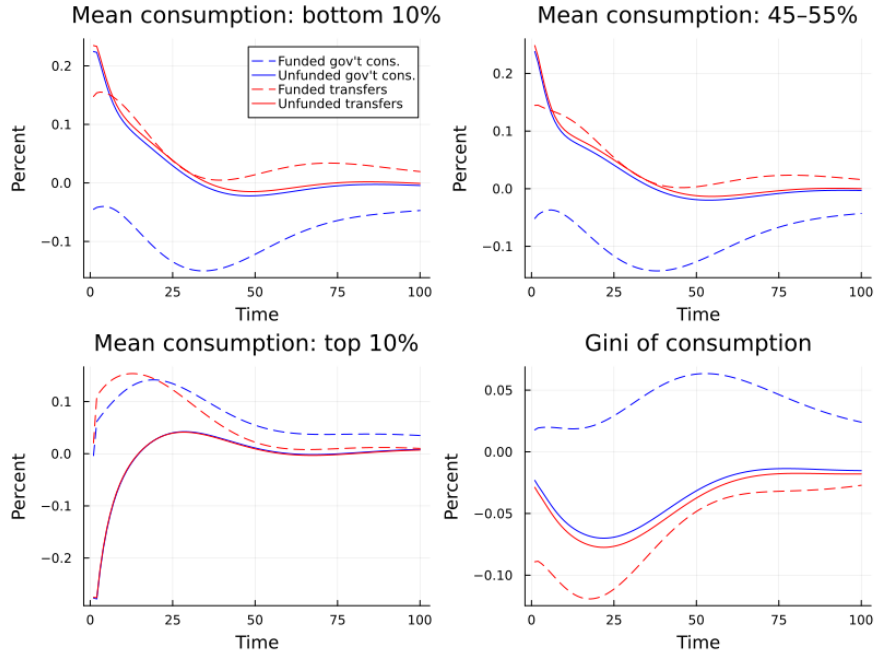
This same mechanism explains why unfunded government-purchases shocks and unfunded transfer shocks look so similar. Once fiscal backing is weak, the common portfolio adjustment away from nominal government bonds becomes the dominant force. Differences between purchases and transfers matter less for aggregate dynamics, so the two unfunded shocks produce nearly identical responses.

By contrast, when the deficit is funded, the source of the deficit matters more. A funded transfer raises disposable income directly, so its effect depends on who receives the transfer and how strongly those households spend. A funded government-purchases shock raises demand because the government directly purchases goods, rather than because households have more disposable income. The associated future fiscal adjustment can crowd out private spending. In the calibration, funded government-purchases shocks are therefore the least expansionary case.

3.2 Distributional Effects: Mean Consumption Across the Liquid-Asset Distribution and Consumption Inequality

Figure 2 reports the impulse responses of mean consumption in the bottom 10 percent, the 45th–55th percentiles, and the top 10 percent of the liquid-asset distribution, together with the consumption Gini. Unfunded government-purchases shocks and unfunded transfer shocks raise mean consumption strongly in the bottom 10 percent and in the 45th–55th percentiles,

Figure 2: Aggregate responses to fiscal deficit shocks: mean consumption across the liquid-asset distribution and consumption inequality



Note: Impulse Response Functions (IRFs) for mean consumption in the bottom 10 percent, the 45th–55th percentiles, and the top 10 percent of the liquid-asset distribution, together with the consumption Gini, following a 0.1-percent temporary increase in fiscal expenditures. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Red lines indicate transfers, and blue lines indicate government consumption. Transfer shocks are distributed evenly across households.

while reducing mean consumption for the top 10 percent. Funded transfers also raise mean consumption for households with low liquid assets. Funded government-purchases shocks lower mean consumption in the bottom 10 percent and in the 45th–55th percentiles, while raising it for the top 10 percent. The consumption Gini falls under both unfunded shocks and under funded transfers, but rises under funded government purchases.

When deficits are unfunded, mean consumption rises strongly in the bottom 10 percent and in the 45th–55th percentiles because aggregate demand and income expand. For households at the top of the liquid-asset distribution, the dominant force is different. Since they hold large positive nominal positions, inflation reduces the real value of those assets and

lowers their purchasing power. Mean consumption therefore falls for the top 10 percent. Consumption inequality therefore declines.

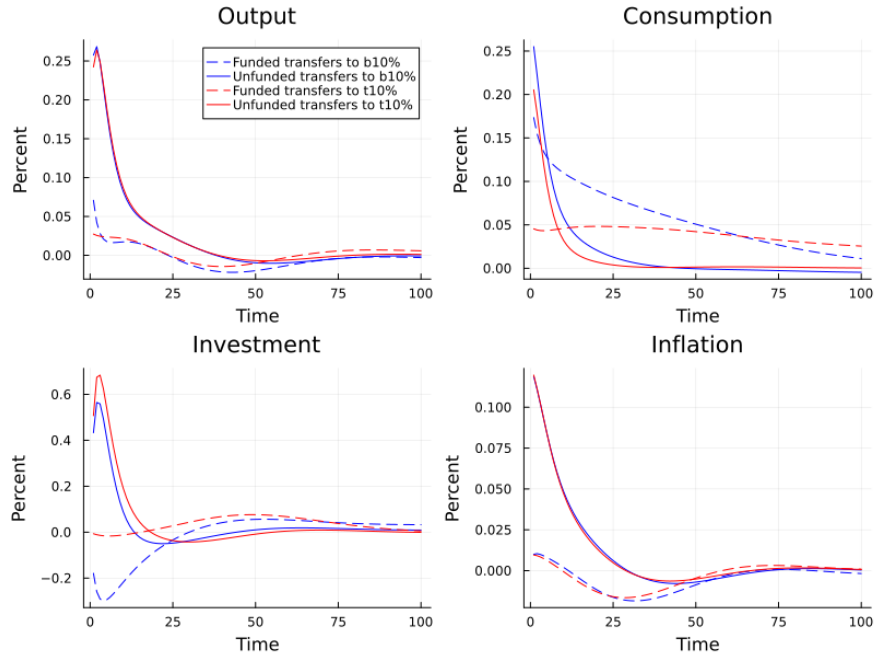
When deficits are funded, the source of the deficit matters more. Funded transfers raise disposable income directly and therefore support households with low liquid assets more strongly. Since these households typically have higher MPCs, their consumption increases. Funded government purchases generate little portfolio reallocation away from nominal government bonds, so their effects work mainly through direct government demand and the implied future fiscal adjustment. In the calibration, this lowers mean consumption in the bottom 10 percent and in the 45th–55th percentiles, while raising it in the top 10 percent. Funded government purchases are therefore the only case in which the consumption Gini rises persistently.

3.3 Impact of Transfer Targeting and Fiscal Backing

Figure 3 reports the impulse responses of output, consumption, investment, and inflation to transfer shocks targeted to either the bottom 10 percent or the top 10 percent of the liquid-asset distribution. In each case, each recipient receives the same amount. Two results stand out. First, when transfers are funded, targeting matters a great deal for consumption and investment. Transfers to the bottom 10 percent generate a much larger rise in consumption than transfers to the top 10 percent. Second, when transfers are unfunded, the difference across recipient groups becomes much smaller for output and inflation. Unfunded transfers to the bottom 10 percent and to the top 10 percent generate very similar responses in those aggregate variables.

When transfers are funded, the main channel is the direct spending response of the recipients. Households at the bottom of the liquid-asset distribution have higher marginal propensities to consume, whereas households at the top have lower marginal propensities to consume. Transfers to the bottom 10 percent therefore raise consumption much more strongly, whereas transfers to the top 10 percent produce a much smaller increase in con-

Figure 3: Aggregate responses to funded and unfunded transfer shocks



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation, showing the effects of a 0.1-percent temporary increase in fiscal transfers. Solid lines represent unfunded shocks, and dashed lines represent funded shocks. Blue lines indicate transfers to the bottom 10 percent of the liquid-asset distribution, and red lines indicate transfers to the top 10 percent. In each case, each recipient receives the same amount.

sumption and a weaker aggregate response.

When transfers are unfunded, fiscal backing adds a second force to this MPC heterogeneity. Households become less willing to hold nominal government bonds at the pre-shock price level, demand shifts toward goods, and inflation rises. That portfolio-reallocation channel pushes up output and inflation even when the transfer is targeted to the top 10 percent. As a result, targeting matters less for those aggregate responses than it does when transfers are funded. At the same time, the consumption response remains stronger when the transfer goes to the bottom 10 percent, because the direct spending channel is still larger for households with low liquid assets.

This comparison shows that targeting and fiscal backing affect transmission in different

ways. Targeting mainly changes the direct spending response of the recipients. Fiscal backing mainly changes whether the transfer also induces a broader shift away from nominal government bonds and into goods. Once the transfer is unfunded, that second channel becomes strong enough that transfers to low- and high-liquid-asset households generate similar movements in output and inflation.

3.4 Role of Portfolio Adjustment Opportunities

This section studies how the funded–unfunded difference depends on households’ ability to rebalance between liquid nominal assets and illiquid capital. The key parameter is the portfolio adjustment probability λ , which governs the fraction of households that can adjust illiquid asset holdings in a given period.

Figure 4 compares impulse responses for two values of λ , 0.005 and 0.03. Lessening the frictions that restrict households’ access to the illiquid capital market enlarges the quantitative gap between the responses to funded and unfunded fiscal deficit shocks: unfunded deficits generate larger responses, while funded deficits generate smaller ones. This pattern is visible for both government-purchases shocks and transfer shocks and is especially clear for output, consumption, and inflation.

A decline in asset-holding frictions has two effects. First, households can self-insure idiosyncratic risk more effectively and smooth consumption more over time. This dampens responses to funded deficits, because households have less need to front-load spending out of additional disposable resources. Second, households can rebalance wealth away from nominal government bonds more easily when fiscal backing weakens. For unfunded deficits, the decline in the real value of nominal government bonds creates a stronger incentive to shift wealth toward illiquid real assets, raising investment and thereby increasing output and inflation.

A higher λ also changes steady-state portfolio composition. When households can adjust illiquid assets more frequently, they rely less on liquid nominal assets for self-insurance and

hold relatively more illiquid wealth. The steady-state liquid-to-illiquid ratio therefore falls. The dynamic responses line up with this portfolio shift.

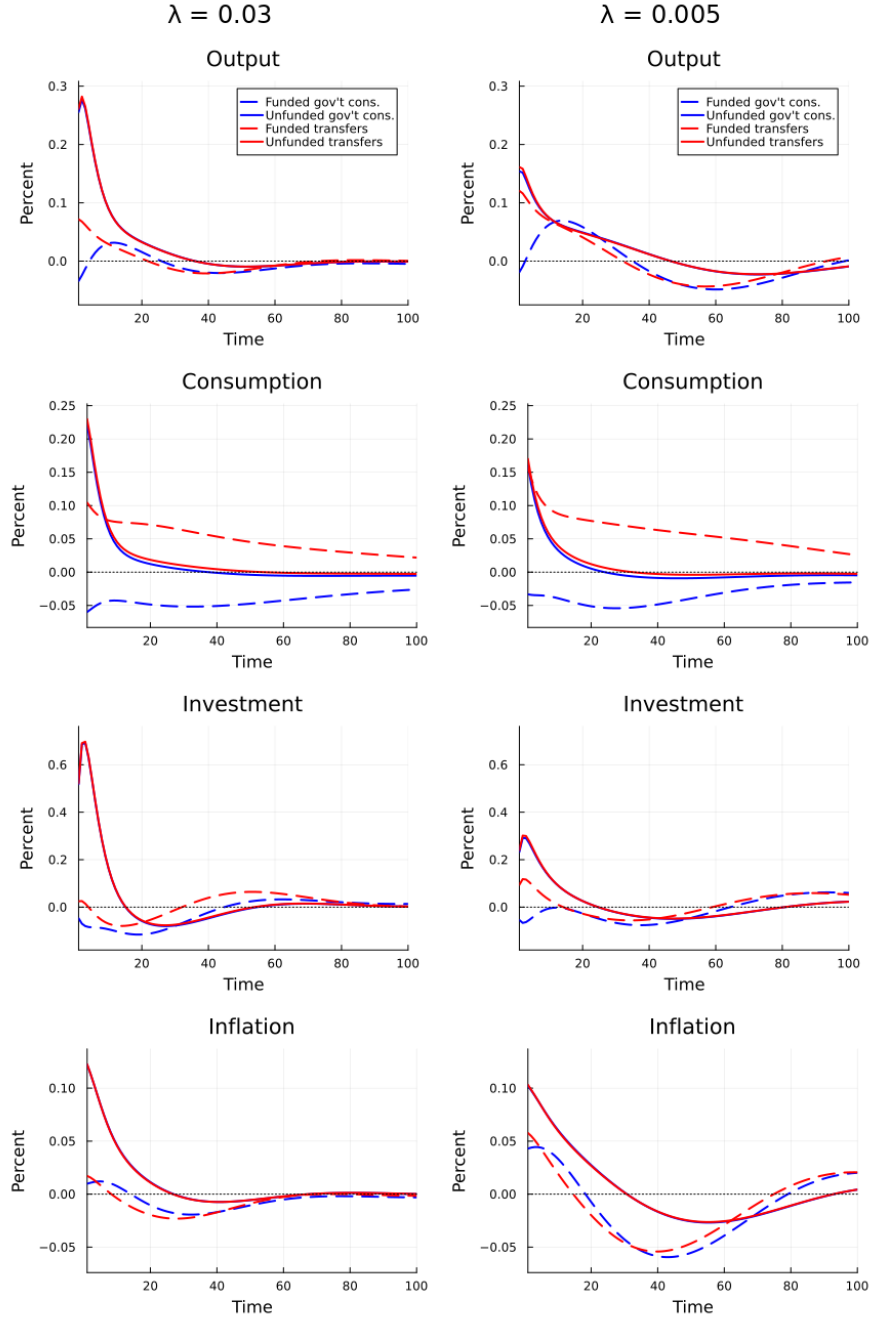


Figure 4: Impulse responses under different portfolio adjustment probabilities. The left column reports responses under $\lambda = 0.03$, and the right column reports responses under $\lambda = 0.005$. Rows show output, consumption, investment, and inflation. A higher λ enlarges the difference between funded and unfunded responses.

These results support the paper’s main mechanism. Household heterogeneity alone does not replace the FTPL explanation of inflation. Once households can reallocate wealth across liquid nominal bonds and illiquid real assets, fiscal backing remains quantitatively central for inflation.

4 Sensitivity to Long-Term Debt Maturity

I also examine sensitivity to the maturity structure of government debt by varying the fraction of long-term debt that matures each period, δ_B . Figure 5 reports the responses to an unfunded transfer shock for $\delta_B = 0.05, 0.2$, and 1. The responses of output, consumption, investment, and inflation are very similar across specifications. A longer debt maturity structure produces slightly smaller effects. Changing debt maturity therefore has little effect on the paper’s main quantitative results.

5 Robustness to Government Spending in Utility

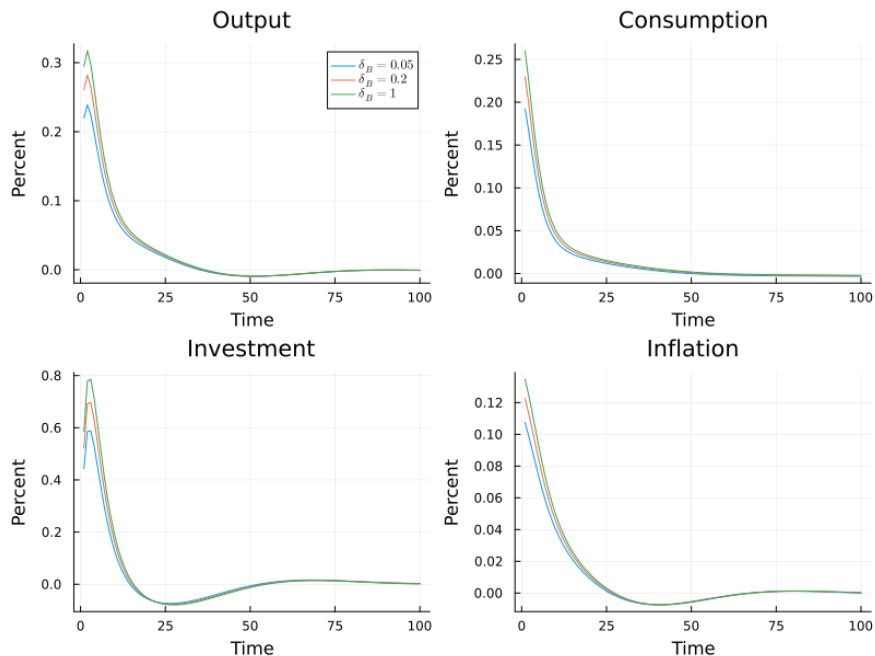
As a robustness check, I allow government spending to enter household utility directly through $c + \alpha_G G$ and vary α_G over a range of values. The main quantitative results are largely unchanged. Across the four fiscal shocks, varying α_G has only small effects on the responses of output, consumption, investment, and inflation. This suggests that the paper’s main conclusions do not depend on excluding government spending from utility.

Appendix D.1 reports the corresponding impulse responses for all four fiscal shocks.

6 Conclusion

This paper studies how fiscal deficits become inflationary in a heterogeneous-agent New Keynesian economy with long-term nominal debt and endogenous household portfolio choice. The key distinction is whether deficits are backed by higher expected future primary sur-

Figure 5: Sensitivity to long-term debt maturity: unfunded transfers



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following an unfunded transfer shock, for alternative values of δ_B , the fraction of long-term debt maturing each period.

pluses. When backing is weak, households reduce desired holdings of nominal government bonds, shift demand toward goods and real assets, and drive up the price level.

The main result is that fiscal backing matters more than the source of the deficit for short-run aggregate dynamics. Unfunded government purchases and unfunded transfers generate very similar responses and are both much more expansionary and inflationary than funded deficits. When deficits are funded, by contrast, inflation and real activity respond much less, and the source of the deficit matters more: funded transfers stimulate private spending more than funded government purchases.

Fiscal backing also changes distributional outcomes. Unfunded deficits reduce the real value of outstanding nominal liabilities, shift purchasing power toward households with higher spending propensities, and reduce consumption inequality. Funded deficits do not generate the same revaluation effect. The portfolio channel further strengthens these re-

sults. When households can adjust illiquid asset positions more easily, the quantitative difference between funded and unfunded deficits becomes larger.

The broader implication is that household heterogeneity does not displace fiscal backing as an explanation for inflation. Once households can reallocate wealth across liquid nominal bonds and illiquid real assets, whether deficits are backed remains quantitatively central for inflation.

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A Additional Model Details

This appendix collects additional details on preferences, taxation, household income, firm-side optimization, and aggregation. The main text focuses on the parts of the model that are central to the paper’s mechanism: funded versus unfunded deficits, household portfolio choice, long-term nominal debt, and the portfolio adjustment friction. This appendix reports the remaining blocks of the quantitative environment.

A.1 Preferences, Labor Supply, and Tax Progressivity

Households have Greenwood–Hercowitz–Huffman (GHH) preferences. Household i chooses consumption c_{it} and labor supply n_{it} to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it} - G(h_{it}, n_{it})), \quad (28)$$

where β is the discount factor, h_{it} is idiosyncratic labor productivity, and $G(h_{it}, n_{it})$ is the disutility of labor. Period utility is of CRRA form,

$$u(x_{it}) = \frac{x_{it}^{1-\xi} - 1}{1-\xi}, \quad (29)$$

where

$$x_{it} = c_{it} - G(h_{it}, n_{it}), \quad (30)$$

and ξ is the risk-aversion parameter.

Following [Benabou \(2002\)](#) and [Heathcote et al. \(2017\)](#), after-tax labor income is

$$y_{it} = (1 - \tau_t^L) (w_t h_{it} n_{it})^{1-\tau_t^P}, \quad (31)$$

where w_t is the aggregate real wage, τ_t^L is the labor tax level, and τ_t^P is tax progressivity. The parameter τ_t^P introduces nonlinear taxation: higher values of τ_t^P imply stronger progressivity.

The marginal disutility of labor is

$$\frac{\partial G(h_{it}, n_{it})}{\partial n_{it}} = (1 - \tau_t^P)(1 - \tau_t^L)(w_t h_{it})^{1-\tau_t^P} n_{it}^{-\tau_t^P} = (1 - \tau_t^P) \frac{y_{it}}{n_{it}}. \quad (32)$$

Assume that the disutility function has constant elasticity with respect to labor supply:

$$\frac{\partial G(h_{it}, n_{it})}{\partial n_{it}} = (1 + \gamma) \frac{G(h_{it}, n_{it})}{n_{it}}, \quad (33)$$

with $\gamma > 0$. Then the composite consumption-labor term can be written as

$$x_{it} = c_{it} - \frac{1 - \tau_t^P}{1 + \gamma} y_{it}. \quad (34)$$

This representation implies that labor disutility is proportional to after-tax labor income. As a result, household decisions can be written in terms of the transformed object x_{it} and after-tax income y_{it} .

For the quantitative solution, I follow [Bayer et al. \(2024\)](#) and assume that tax progressivity is constant at its equilibrium value in the labor-supply block, $\tau_t^P \equiv \bar{\tau}^P$, and that

$$G(h_{it}, n_{it}) = h_{it}^{1-\bar{\tau}^P} \frac{n_{it}^{1+\gamma}}{1 + \gamma}. \quad (35)$$

Under this specification, the first-order condition for labor supply implies that individual productivity does not affect the optimal labor choice. All employed households therefore choose the same labor supply, denoted by N_t .

Given the normalization $\int h_{it} di \equiv 1$, total effective labor input is

$$\int h_{it} n_{it} di = N_t. \quad (36)$$

Substituting the common labor choice into the labor-income expression yields

$$y_{it} = (1 - \tau_t^L)(w_t h_{it} n_{it})^{1-\bar{\tau}^P} \quad (37)$$

$$= (1 - \tau_t^L)^{\frac{1+\gamma}{\gamma+\bar{\tau}^P}} (1 - \bar{\tau}^P)^{\frac{1-\bar{\tau}^P}{\gamma+\bar{\tau}^P}} w_t^{\frac{1+\gamma}{\gamma+\bar{\tau}^P} (1-\bar{\tau}^P)} h_{it}^{1-\bar{\tau}^P}. \quad (38)$$

Income heterogeneity therefore arises from heterogeneity in labor productivity.

A.2 Household Income, Occupation, and Profit Distribution

The household block follows [Bayer et al. \(2024\)](#). I distinguish between workers and entrepreneurs. Workers face idiosyncratic labor-income risk. Entrepreneurs correspond to the zero-labor state and receive non-union firm profits.

Let $\tilde{h}_{it}$ denote pre-normalized productivity. Productivity evolves as

$$\tilde{h}_{it} = \begin{cases} \exp(\rho_h \log \tilde{h}_{it-1} + \epsilon_{it}^h) & \text{with probability } 1 - \zeta \text{ if } h_{it-1} \neq 0, \\ 1 & \text{with probability } \iota \text{ if } h_{it-1} = 0, \\ 0 & \text{otherwise,} \end{cases} \quad (39)$$

where

$$h_{it} = \frac{\tilde{h}_{it}}{\int \tilde{h}_{it} di}. \quad (40)$$

Households with $h_{it} \neq 0$ are workers, while households with $h_{it} = 0$ are entrepreneurs. Worker productivity follows a log-AR(1) process with persistence ρ_h and innovation $\epsilon_{it}^h \sim \mathcal{N}(0, \bar{\sigma}_h^2)$. The parameter ζ governs the transition from worker to entrepreneur, and ι governs the transition from entrepreneur to worker.

Workers receive labor income and a lump-sum share of union profits. Entrepreneurs receive firm profits other than union profits. Entrepreneurs can liquidate only a fraction ω^Π of firm profits through tradable profit claims with stochastic maturity. A fraction ι^Π of existing claims matures each period. Let q_t^Π denote the price of profit-claim paper. Entrepreneurial profit income is then

$$\Pi_t^E = (1 - \omega^\Pi)\Pi_t^F + \iota^\Pi q_t^\Pi, \quad (41)$$

where Π_t^F denotes firm profits.

Substituting the transformed variable

$$c_{it} = x_{it} + \frac{1 - \bar{\tau}^P}{1 + \gamma} y_{it} \quad (42)$$

into the household budget constraint gives

$$\begin{aligned} x_{it} + b_{it+1} + q_t k_{it+1} &= \frac{R_{it}}{\pi_t} b_{it} + (q_t + r_t) k_{it} + \frac{\bar{\tau}^P + \gamma}{1 + \gamma} y_{it} \\ &+ \mathbb{I}_{h_{it} \neq 0} (1 - \tau_t) \Pi_t^U + \mathbb{I}_{h_{it} = 0} (1 - \tau_t^L) (\Pi_t^E)^{1 - \bar{\tau}^P} + Transfers_{it}, \end{aligned} \quad (43)$$

subject to

$$k_{it+1} \geq 0, \quad b_{it+1} \geq \underline{B}. \quad (44)$$

A.3 Tax Revenue Construction

The average tax rate τ_t in the main text is the ratio of total tax revenue to total pre-tax labor and entrepreneurial income. Let w_t denote the real wage, n_{it} hours worked, and h_{it} idiosyncratic productivity. The average tax rate is

$$\tau_t = \frac{\mathbb{E}_t[w_t n_{it} h_{it} + \mathbb{I}_{h_{it} = 0} \Pi_t^E] - \tau_t^L \mathbb{E}_t[(w_t h_{it} n_{it} + \mathbb{I}_{h_{it} = 0} \Pi_t^E)^{1 - \tau_t^P}]}{\mathbb{E}_t[w_t n_{it} h_{it} + \mathbb{I}_{h_{it} = 0} \Pi_t^E]}. \quad (45)$$

The level parameter τ_t^L adjusts to match the targeted average tax rate implied by the tax rule in the main text.

Aggregate tax revenue is then

$$Tax_t = \tau_t \mathbb{E}_t[w_t n_{it} h_{it} + \mathbb{I}_{h_{it} \neq 0} \Pi_t^U + \mathbb{I}_{h_{it} = 0} \Pi_t^E], \quad (46)$$

where Π_t^U denotes union profits and Π_t^E denotes entrepreneurial profits.

A.4 Firm-Side Problem

This section reports the full firm-side block underlying the reduced-form description in the main text.

A.4.1 Final Goods Producers

Final goods producers aggregate a continuum of differentiated intermediate goods y_{jt} into final output Y_t according to a CES aggregator:

$$Y_t = \left(\int y_{jt}^{\frac{\eta_t-1}{\eta_t}} dj \right)^{\frac{\eta_t}{\eta_t-1}}, \quad (47)$$

where η_t is the elasticity of substitution across varieties. The associated price index is

$$P_t = \left(\int p_{jt}^{1-\eta_t} dj \right)^{\frac{1}{1-\eta_t}}, \quad (48)$$

and demand for variety j is

$$y_{jt} = \left(\frac{p_{jt}}{P_t} \right)^{-\eta_t} Y_t. \quad (49)$$

A.4.2 Intermediate Goods Producers

Intermediate goods producers operate under monopolistic competition and produce with Cobb–Douglas technology:

$$y_{jt} = Z_t N_{jt}^\alpha (u_{jt} K_{jt})^{1-\alpha}, \quad (50)$$

where Z_t is total factor productivity, N_{jt} is labor, K_{jt} is capital, and u_{jt} is utilization.

Capital depreciation increases with utilization:

$$\delta(u_{jt}) = \delta_0 + \delta_1(u_{jt} - 1) + \frac{\delta_2}{2}(u_{jt} - 1)^2. \quad (51)$$

Firm j minimizes cost

$$w_t^F N_{jt} + (r_t + q_t \delta(u_{jt})) K_{jt},$$

which yields the first-order conditions

$$w_t^F = \alpha m c_{jt} Z_t \left(\frac{u_{jt} K_{jt}}{N_{jt}} \right)^{1-\alpha}, \quad (52)$$

$$r_t + q_t \delta(u_{jt}) = u_{jt}(1 - \alpha) mc_{jt} Z_t \left(\frac{N_{jt}}{u_{jt} K_{jt}} \right)^\alpha, \quad (53)$$

and

$$q_t [\delta_1 + \delta_2(u_{jt} - 1)] = (1 - \alpha) mc_{jt} Z_t \left(\frac{N_{jt}}{u_{jt} K_{jt}} \right)^\alpha. \quad (54)$$

Under symmetry, marginal costs are equalized across firms: $mc_{jt} = mc_t$.

Price setting is subject to Calvo frictions. Each period, a firm can reset its price with probability $1 - \lambda_Y$. The firm maximizes the expected discounted value of profits:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_Y^t (1 - \tau_t^L) Y_t^{1 - \tau_t^P} \left\{ \left(\frac{p_{jt} \bar{\pi}^t}{P_t} - mc_t \right) \left(\frac{p_{jt} \bar{\pi}^t}{P_t} \right)^{-\eta_t} \right\}^{1 - \tau_t^P}. \quad (55)$$

The corresponding price Phillips curve is

$$\log \left(\frac{\pi_t}{\bar{\pi}} \right) = \beta \mathbb{E}_t \log \left(\frac{\pi_{t+1}}{\bar{\pi}} \right) + \kappa_Y \left(mc_t - \frac{1}{\mu_t^Y} \right), \quad (56)$$

where

$$\kappa_Y = \frac{(1 - \lambda_Y)(1 - \beta \lambda_Y)}{\lambda_Y}, \quad \frac{1}{\mu_t^Y} = \frac{\eta_t - 1}{\eta_t}. \quad (57)$$

A.4.3 Capital Goods Producers

Capital goods producers take the price of capital q_t as given and choose investment I_t to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t I_t \left\{ \Psi_t q_t \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] - 1 \right\}, \quad (58)$$

where Ψ_t is the marginal efficiency of investment and ϕ governs adjustment costs.

The optimality condition for investment is

$$\Psi_t q_t \left[1 - \phi \log \left(\frac{I_t}{I_{t-1}} \right) \right] = 1 - \beta \mathbb{E}_t \left[\Psi_{t+1} q_{t+1} \phi \log \left(\frac{I_{t+1}}{I_t} \right) \right]. \quad (59)$$

The law of motion for capital is

$$K_t = (1 - \delta(u_t))K_{t-1} + \Psi_t \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] I_t. \quad (60)$$

A.4.4 Labor Packers and Unions

Households sell labor to unions, which supply differentiated labor services to labor packers.

Labor packers aggregate labor services into total labor input:

$$N_t = \left(\int \hat{n}_{jt}^{\frac{\zeta_t-1}{\zeta_t}} dj \right)^{\frac{\zeta_t}{\zeta_t-1}}, \quad (61)$$

where $\zeta_t > 1$ is the elasticity of substitution across labor types. Demand for labor type j is

$$\hat{n}_{jt} = \left(\frac{W_{jt}}{W_t^F} \right)^{-\zeta_t} N_t, \quad (62)$$

where W_{jt} is the nominal wage set by union j and W_t^F is the nominal wage paid by labor packers.

Unions face Calvo wage rigidity. Each period, a union can reset its wage with probability $1 - \lambda_w$. When reoptimizing, a union maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_w \frac{W_t^F}{P_t} N_t \left\{ \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} - \frac{W_t}{W_t^F} \right) \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} \right)^{-\zeta_t} \right\}. \quad (63)$$

The wage Phillips curve is

$$\log \left(\frac{\pi_t^W}{\bar{\pi}_W} \right) = \beta \mathbb{E}_t \log \left(\frac{\pi_{t+1}^W}{\bar{\pi}_W} \right) + \kappa_w \left(mc_t^w - \frac{1}{\mu_t^W} \right), \quad (64)$$

where

$$\pi_t^W = \frac{W_t^F}{W_{t-1}^F} = \frac{w_t^F}{w_{t-1}^F} \pi_t, \quad mc_t^w = \frac{w_t}{w_t^F}, \quad \frac{1}{\mu_t^W} = \frac{\xi_t - 1}{\xi_t}, \quad (65)$$

and

$$\kappa_w = \frac{(1 - \lambda_w)(1 - \beta\lambda_w)}{\lambda_w}. \quad (66)$$

A.5 Additional Aggregation and Market-Clearing Details

The liquid asset market clears when aggregate household demand for liquid positions equals the supply of liquid claims backed by government debt and tradable profit claims:

$$B_{t+1} = \mathbb{E}_t [\lambda b_{a,t}^* + (1 - \lambda) b_{n,t}^*]. \quad (67)$$

In equilibrium, the supply of liquid claims is supported by the long-term government bond portfolio intermediated by banks and by profit-claim paper.

The price of profit-claim paper satisfies the no-arbitrage condition

$$q_t^\Pi R_t^b = \mathbb{E}_t \pi_{t+1} [(1 - \iota^\Pi) q_{t+1}^\Pi + \omega^\Pi \Pi_{t+1}^F]. \quad (68)$$

The capital market clears when aggregate household demand for illiquid capital equals the aggregate capital stock:

$$K_{t+1} = \mathbb{E}_t [\lambda k_t^* + (1 - \lambda) k]. \quad (69)$$

The labor market clears at the equilibrium real wage, and the goods market clears through the resource constraint. Together, the liquid asset market, capital market, labor market, and goods market determine equilibrium prices and quantities.

B Additional Calibration Details

This appendix reports the full steady-state calibration, implementation constants, policy-rule parameters, shock processes, and additional remarks on the parameters most closely tied to

the paper’s mechanism. The main text highlights only the parameters most closely tied to household portfolio choice, long-term nominal debt, and the fiscal rules that distinguish funded from unfunded deficits.

B.1 Steady-State Calibration

I pin down the steady state by fixing a set of preference, technology, income-process, liquidity/portfolio, and fiscal parameters so that the model matches the same broad quarterly steady-state targets as [Bayer et al. \(2024\)](#). The targeted objects include earnings persistence and dispersion, worker–entrepreneur flows, the liquid–illiquid composition of household wealth, borrower prevalence, factor shares, government size, and tax progressivity.

Table 3: Steady-state calibration: parameters and targets (quarterly)

Block	Symbol	Value	Target / Mapping	Data / Source
<i>Preferences and labor supply</i>				
Risk aversion	ξ	4.000	Precautionary saving strength	Kaplan–Violante (2014)
Inverse Frisch elasticity	γ	2.000	Labor supply slope	Chetty et al. (2011)
Discount factor	β	0.98255	K/Y	NIPA long-run averages
Labor disutility scale	γ_{scale}	0.200	Steady hours in [0.3, 0.4]	BLS/NIPA (hours)
<i>Income process and occupation</i>				
Persistence (labor inc.)	ρ_h	0.980	Earnings persistence	Storesletten–Telmer–Yaron (2004)
Std. dev. (labor inc.)	σ_h	0.120	Earnings dispersion	Storesletten–Telmer–Yaron (2004)

Continued on next page

Table 3: Steady-state calibration: parameters and targets (quarterly)
(*continued*)

Block	Symbol	Value	Target / Mapping	Data / Source
$E \rightarrow W$ transition	ι	$1/16 = 0.0625$	Worker share in steady state	Guvenen–Kaplan–Song (2014)
$W \rightarrow E$ transition	ζ	$1/4500 \approx 2.22 \times 10^{-4}$	Upper-tail wealth concentration	WID/SCF
<i>Technology and markups</i>				
Capital share	α	0.318	Labor share ≈ 0.68	NIPA factor shares
Depreciation (quarterly)	δ_0	0.0215	K/Y	NIPA CFC
Price markup (gross)	μ^Y	1.10	Micro price-cost margins	Standard benchmark
Wage markup (gross)	μ^W	1.10	Micro wage markup	Standard benchmark
<i>Liquidity, maturity, and asset supply</i>				
Portfolio adj. prob.	λ	0.030	Liquid/illiquid turnover	SCF/NIPA (as in BBL)
Bond maturity	δ_B	0.200	Avg. duration $= 1/\delta_B$	Treasury duration stylized fact
Bonds-to-capital	ψ	0.100	$(B + \bar{q}_\Pi)/K$	SCF/NIPA aggregate ratio
Tradable profit share	ω_Π	0.150	Liquid equity-like paper	BBL steady-state mapping
Retirement of profit shares	ι_Π	0.016	Liquid claim turnover	BBL steady-state mapping
<i>Fiscal block and normalizations</i>				

Continued on next page

Table 3: Steady-state calibration: parameters and targets (quarterly)
(*continued*)

Block	Symbol	Value	Target / Mapping	Data / Source
$1 - \tau^L$	$1 - \tau^L$	0.645	$G/Y \approx 0.20$ in steady state	NIPA government share
Tax progressivity	τ^P	0.1022	Average progressivity	SOI
Gross inflation	π	1.000	Normalization (indexation)	Convention
Gross policy rate	R^b	1.000	Normalization (quarterly)	Convention
Borrower wedge	$\bar{R}$	0.0466	Borrower share $\sim 16\%$	SCF (as in BBL)
Return on capital	R	1.010	Scale consistency with β, δ_0	Normalization

Notes: The steady-state calibration targets the same broad long-run empirical objects as [Bayer et al. \(2024\)](#): income persistence and dispersion, worker–entrepreneur flows, the combined liquid paper supply ($B + \bar{q}_\Pi$) relative to illiquid capital K , the prevalence of borrowers, factor shares, government size, and tax progressivity. “BBL steady-state mapping” refers to the accounting identity linking (ω_Π, ι_Π) to $\bar{q}_\Pi/Y$ in their steady state. Inflation and the gross policy rate are normalized under indexation. The borrower wedge is implied by $\bar{R} = \pi(1.20)^{1/4} - 1$ in the code.

B.2 Implementation Constants and Scenario Controls

A small number of parameters act as scenario toggles or implementation constants. These objects do not discipline the steady-state targets in Table 3, but they are part of the quantitative implementation.

B.3 Policy Rules and Shock Processes

This section reports the full set of dynamic policy-rule parameters and exogenous shock processes used in the benchmark simulations. The funded and unfunded fiscal rules are the main policy objects relevant for the paper’s mechanism. The remaining shocks are included

Table 4: Implementation constants and scenario controls

Symbol	Value	Role	Comment
ASHIFT	1.000	Borrowing-wedge shift	Implementation scale
$\omega_{\Pi}^{\text{bar}}$	0.20	Profit-claim scaling	Complements $(\omega_{\Pi}, \iota_{\Pi})$ mapping
TRANSFERtoG	1.00	Baseline Transfer/ G ratio	Scenario toggle
BFRatio, GFRatio	0.95, 0.95	Tax / adjustment shares	Fiscal decomposition
α_G	0.0	Government spending in utility	Scenario toggle

to complete the quantitative environment.

Table 5: Policy-rule and shock-process parameters (calibrated baseline)

Block	Symbol	Value	Symbol	Value	Symbol	Value
<i>Exogenous aggregate shocks</i>						
Spread / wedge	ρ_A	0.90	σ_A	0.001		
TFP	ρ_Z	0.90	σ_Z	0.001		
Investment-good productivity	ρ_{Ψ}	0.90	σ_{Ψ}	0.001		
Price markup	ρ_{μ^Y}	0.90	σ_{μ^Y}	0.001		
Wage markup	ρ_{μ^W}	0.90	σ_{μ^W}	0.001		
<i>Income-risk process</i>						
Income-risk state	ρ_s	0.84	σ_s	0.001		
Cyclical sensitivity	Σ_N	0.00				
<i>Monetary policy</i>						
Taylor rule	ρ_R	0.70	θ_{π}	0.50	θ_Y	0.125
Monetary shock	σ_R	0.001				
<i>Fiscal policy: tax rules</i>						
Avg. tax rate	ρ_{τ}	0.50	γ_B^{τ}	0.00	γ_Y^{τ}	0.00
Progressivity	ρ_P	0.50	γ_B^P	0.00	γ_Y^P	0.00

Continued on next page

Table 5: Policy-rule and shock-process parameters (calibrated baseline) (*continued*)

Block	Symbol	Value	Symbol	Value	Symbol	Value
Progressivity shock	σ_P	0.001				
<i>Fiscal components: purchases and transfers</i>						
GF shock	$\rho_{GFshock}$	0.98	$\sigma_{GFshock}$	0.001		
GU shock	$\rho_{GUshock}$	0.98	$\sigma_{GUshock}$	0.001		
TF shock	$\rho_{TFshock}$	0.98	$\sigma_{TFshock}$	0.001		
TU shock	$\rho_{TUshock}$	0.98	$\sigma_{TUshock}$	0.001		
GF rule	ρ_{GF}	0.90	γ_{YGF}	0.50	γ_{BGF}	-0.50
GU rule	ρ_{GU}	0.90	γ_{YGU}	0.50	γ_{BGU}	0.50
TF rule	ρ_{TF}	0.90	γ_{YTF}	0.50	γ_{BTF}	-0.50
TU rule	ρ_{TU}	0.90	γ_{YTU}	0.50	γ_{BTU}	0.50
<i>Notes:</i> Values report the calibrated baseline used in the simulations. Government purchases and transfers are decomposed as $G_t = GU_t + GF_t$ and $Transfer_t = TU_t + TF_t$. Total long-term nominal government debt is decomposed as $Bgov_t = BFgov_t + BUGov_t$, where $BUGov_t \equiv Bgov_t - BFgov_t$. The rules for GF_t and TF_t feed back on $BFgov_t$, while the rules for GU_t and TU_t feed back on $BUGov_t$. Negative coefficients on funded debt imply stabilizing future adjustment, while nonnegative coefficients on unfunded debt imply the absence of debt-stabilizing backing.						

B.4 Calibration Objects Relevant for the Mechanism

Four sets of objects are especially important for the paper's mechanism. First, the portfolio adjustment probability λ governs how easily households can rebalance between liquid nominal government liabilities and illiquid real capital. Second, the maturity parameter δ_B governs the duration of government debt and therefore the strength of nominal valuation

effects. Third, the liquid-asset supply parameters $(\psi, \omega_\Pi, \iota_\Pi)$ determine households' exposure to liquid nominal claims. Fourth, the fiscal feedback coefficients $(\gamma_{BGF}, \gamma_{BTF}, \gamma_{BGU}, \gamma_{BTU})$ determine whether a deficit is funded or unfunded in equilibrium.

The quantitative exercises in the paper vary some of these objects to assess mechanism strength. In particular, changes in λ alter households' ability to rebalance across assets and therefore provide a within-model test of the role of portfolio adjustment in amplifying the gap between funded and unfunded deficits.

C Numerical Solution Methods

This appendix documents the numerical solution of the two-asset heterogeneous-agent New Keynesian model with portfolio adjustment frictions and long-term nominal government debt. The implementation follows a *Reiter-style* strategy: (i) compute a stationary equilibrium by jointly solving household problems and the invariant distribution; (ii) construct a reduced-dimensional state representation of the cross-sectional distribution; (iii) linearize the full equilibrium system around the stationary equilibrium; and (iv) solve the resulting linear rational-expectations system via generalized Schur decomposition to obtain policy functions and impulse responses.

C.1 Overview of the computational approach

Let the individual state be (m, k, s) , where m denotes bond positions (deposits/bonds, allowing for borrowing), k denotes illiquid real capital, and s collects idiosyncratic income/occupation states (worker productivity states plus an entrepreneur state). Aggregate states include nominal rigidities and policy variables (e.g., markups, monetary policy, fiscal components, taxes), and the cross-sectional distribution $\Theta_t(m, k, s)$.

The computation proceeds in four layers:

1. **Discretization.** Construct grids for (m, k) and a Markov chain for s .

2. **Household block.** Solve two Bellman problems each period: (i) *adjusters* (can choose both m' and k') and (ii) *non-adjusters* (can choose m' only; k is predetermined). The solution uses an endogenous-grid method (EGM) combined with an upper-envelope step for the discrete k' choice.
3. **Distribution and aggregation.** Given policy functions, iterate the law of motion forward using a mass-preserving linear interpolation scheme to obtain the stationary distribution and aggregates.
4. **Linearization and dynamics.** Reduce the dimensionality of Θ_t , compute Jacobians of the equilibrium conditions, solve the linear system with QZ/Schur methods, and generate IRFs.

C.2 Discretization: idiosyncratic risk and asset grids

Idiosyncratic states. Worker productivity follows a log-AR(1) process. I approximate it by a finite-state Markov chain with n_s states using a Tauchen-style method. Occupation switching between workers and entrepreneurs is embedded by augmenting the state space with an entrepreneur state and modifying the transition matrix accordingly. Let $\Phi \in \mathbb{R}^{n_s \times n_s}$ denote the resulting transition matrix.

Liquid and illiquid grids. I discretize liquid assets m on a non-uniform grid that is dense near the borrowing constraint $\underline{m} < 0$ to resolve kinks at the constraint and high MPC regions. Illiquid capital k is discretized on a non-uniform grid that is dense at low wealth to capture participation/adjustment behavior and sparse at the top to cover the tail efficiently. Denote grid sizes by n_m and n_k .

Portfolio adjustment. Each period, a fraction λ of households can adjust illiquid capital. With probability $1 - \lambda$, the household keeps $k' = k$ and only chooses m' . This structure

yields two policy functions for liquid savings: $m'_a(m, k, s)$ for adjusters and $m'_n(m, k, s)$ for non-adjusters, plus one capital policy $k'(m, k, s)$ for adjusters.

C.3 Household problem and EGM implementation

C.3.1 Recursive formulation

Let $W_t(m, k, s)$ denote the continuation value *before* the capital-adjustment draw. The two conditional value functions are

$$V_t^a(m, k, s) = \max_{m', k'} u(x_t) + \beta \mathbb{E} [W_{t+1}(m', k', s') \mid s], \quad (70)$$

$$V_t^n(m, k, s) = \max_{m'} u(x_t) + \beta \mathbb{E} [W_{t+1}(m', k, s') \mid s], \quad (71)$$

$$W_t(m, k, s) = \lambda V_t^a(m, k, s) + (1 - \lambda) V_t^n(m, k, s), \quad (72)$$

subject to the budget constraint and constraints on (m', k') . The composite consumption term x_t is the control entering the GHH utility specification used in the main text.

C.3.2 Euler equations and the EGM step

The liquid Euler equation takes the form

$$u_x(x_t) = \beta \mathbb{E}_t [u_x(x_{t+1}) R_{t+1}^m], \quad (73)$$

where R_{t+1}^m is the relevant real gross return on bond positions (accounting for deposits/bonds, inflation, and the borrower wedge when $m < 0$).

For a given candidate next-period state (m', k', s') , define the expected marginal value of liquid wealth

$$\mathcal{M}_t(m', k', s) := \mathbb{E} [u_x(x_{t+1}) R_{t+1}^m \mid s]. \quad (74)$$

Then (73) implies

$$u_x(x_t) = \beta \mathcal{M}_t(m', k', s) \quad \Rightarrow \quad x_t = u_x^{-1}(\beta \mathcal{M}_t(m', k', s)). \quad (75)$$

Non-adjusters. For non-adjusters, $k' = k$ is fixed. The EGM constructs an *endogenous* current-state grid for m by combining (75) with the one-period budget constraint:

$$m = \frac{m' + q_t k' + x_t - \text{income}_t(m, k, s)}{R_t^m}, \quad (76)$$

where $\text{income}_t(\cdot)$ collects labor/profit income, transfers, and returns on k at given prices. The resulting mapping ($m' \mapsto m$) is monotone except at the borrowing constraint; I enforce $m' \geq \underline{m}$ and apply the standard EGM correction at the constraint (consumption becomes residual).

Adjusters. For adjusters, the household chooses both m' and k' . I implement a two-asset EGM in the spirit of “grid search over k' + EGM over m' ”:

1. Fix a candidate k' on the illiquid grid.
2. For this k' , apply the liquid EGM step exactly as for non-adjusters, constructing the implied x_t and endogenous m grid for each (m', s) .
3. Compute the implied value for each k' and take the *upper envelope* across k' :

$$V_t^a(m, k, s) = \max_{k' \in \mathcal{K}} \tilde{V}_t^a(m, k, s; k').$$

The upper-envelope step is essential because non-concavities can arise from the discrete capital-adjustment opportunity and from the borrowing constraint in m .

Interpolation. The EGM step yields policies on an endogenous grid. I map policies back to the exogenous (m, k) grid using monotone piecewise-linear interpolation. This preserves

feasibility and avoids spurious oscillations near kinks.

C.4 Forward iteration for the cross-sectional distribution

Given policy functions and the idiosyncratic transition matrix Φ , the distribution evolves via a discrete-time Kolmogorov forward equation. Let Θ_t denote the probability mass on the tensor grid $\mathcal{M} \times \mathcal{K} \times \mathcal{S}$. The next-period distribution is constructed as:

1. For each current grid point (m, k, s) , compute the policy-implied next states:

$$(m'_a, k') \text{ if adjusting,} \quad (m'_n, k) \text{ if not adjusting.}$$

2. Split mass using a mass-preserving linear interpolation scheme: if m' (or k') lies between adjacent grid points, allocate weights proportional to distance so that aggregate mass is preserved and the implied law of motion is linear in Θ_t .
3. Apply idiosyncratic transitions: for each origin state s , distribute mass across s' using $\Phi(s, s')$.
4. Combine adjusters and non-adjusters:

$$\Theta_{t+1} = \lambda \mathcal{T}_a(\Theta_t) + (1 - \lambda) \mathcal{T}_n(\Theta_t),$$

where $\mathcal{T}_a$ and $\mathcal{T}_n$ are the implied transition operators.

Stationary distribution. The invariant distribution $\bar{\Theta}$ solves $\bar{\Theta} = \mathcal{T}(\bar{\Theta})$. I compute it by forward iteration with damping until $\|\Theta^{(j+1)} - \Theta^{(j)}\|_\infty$ falls below a tight tolerance and the distribution sums to one up to machine precision.

C.5 Stationary equilibrium (steady state) computation

A stationary equilibrium consists of prices, policy variables, household policies, and an invariant distribution that jointly satisfy: (i) household optimality, (ii) market clearing in liquid assets, capital, labor, and goods, (iii) government budget identities and fiscal rules in steady state, and (iv) pricing and wage-setting steady-state restrictions.

Outer loop. I solve for a vector of steady-state unknowns (e.g., $(q, w, r, \tau, \dots)$, and objects implied by the long-term bond pricing relation) using a damped fixed-point or Broyden routine:

1. Guess the steady-state price and policy vector.
2. Solve the household block at those prices.
3. Compute $\bar{\Theta}$ by forward iteration.
4. Aggregate and evaluate market-clearing residuals.
5. Update the guess using a quasi-Newton step with damping.

Long-term nominal bond pricing. Long-term government debt has geometric maturity δ_B . In steady state, the no-arbitrage condition implies a closed-form price:

$$\bar{q}^B = \frac{1}{\bar{R}^b - (1 - \delta_B)}. \quad (77)$$

I use this identity as an additional numerical check and as a stabilization device inside the steady-state routine.

C.6 Distribution reduction for linear dynamics

A full tensor distribution $\Theta_t \in \mathbb{R}^{n_m n_k n_s}$ is too large to include as a state vector in the linear system. I therefore compress the distribution using a frequency-domain representation based

on a discrete cosine transform (DCT).

DCT representation. Let θ denote a vectorized object defined on a grid of length n . Let $D \in \mathbb{R}^{n \times n}$ be the DCT matrix. The DCT coefficients are

$$\hat{\theta} = D\theta.$$

Low-frequency coefficients capture smooth variation, while high-frequency coefficients capture local curvature and kinks.

Energy-based coefficient selection. I select a subset of coefficients that retains a $(1 - \text{reduc_value})$ share of the squared ℓ_2 norm of the transformed object:

$$\frac{\|\hat{\theta}_{\mathcal{I}}\|_2}{\|\hat{\theta}\|_2} \geq 1 - \text{reduc_value}.$$

The index set $\mathcal{I}$ is constructed by sorting coefficients by magnitude and adding coefficients until the criterion holds. This produces a sparse representation with minimal loss in ℓ_2 accuracy.

Reduced state vector. Let S denote the selection matrix that maps full DCT coefficients into the retained subset: $\tilde{\theta} = S\hat{\theta}$. In linear dynamics, the cross-sectional state is replaced by $\tilde{\theta}$ together with aggregate states, which makes the Schur solve feasible while preserving distributional variation relevant for aggregates and inequality.

Reconstruction and normalization. When reconstructing objects from reduced coefficients, I apply the inverse DCT and renormalize probability objects so that they integrate to one and remain nonnegative up to numerical tolerance. When needed, I convert PDFs to CDFs by a stable trapezoidal integration routine to avoid numerical drift.

C.7 Linearization and solution of the rational-expectations system

Equilibrium system. Stack aggregate equilibrium conditions and reduced distribution laws into

$$\mathbb{E}_t [F(Z_{t+1}, Z_t, \varepsilon_{t+1})] = 0, \quad (78)$$

where Z_t stacks aggregate states, aggregate controls, and reduced distribution coefficients.

Jacobians. I compute the Jacobians of F at the steady state using automatic differentiation when the mapping is smooth, and otherwise using two-sided finite differences with carefully chosen step sizes. This yields matrices (A, B, C) such that

$$A \Delta Z_{t+1} = B \Delta Z_t + C \varepsilon_{t+1}. \quad (79)$$

Schur/QZ solution. I solve (79) via generalized Schur decomposition. Let the QZ factorization sort eigenvalues so that stable roots lie inside the unit circle. The resulting policy functions take the standard form:

$$\Delta Z_{t+1} = H \Delta Z_t + M \varepsilon_{t+1}, \quad \Delta X_t = G \Delta Z_t,$$

where ΔX_t denotes selected controls. I verify determinacy by checking that the number of stable eigenvalues matches the number of predetermined states.

C.8 Impulse responses

Linear IRFs. Given (H, M) , IRFs are computed by iterating forward from an initial innovation ε_1 :

$$\Delta Z_{t+1} = H \Delta Z_t, \quad t \geq 1, \quad \Delta Z_1 = M \varepsilon_1.$$

All reported IRFs are scaled to a common fiscal innovation size, and I report both aggregate series and distributional objects implied by the reduced state.

C.9 Diagnostics and numerical accuracy

I monitor the following accuracy checks:

1. **Euler residuals.** Sup-norm and percentile statistics of the liquid Euler equation error in consumption units, separately for borrowers and savers.
2. **Market clearing.** Residuals in the liquid asset market, capital market, and government budget identities.
3. **Mass conservation.** $\sum \Theta_t = 1$ each iteration and non-negativity of Θ_t up to numerical tolerance.
4. **Grid sensitivity.** Key aggregates and multipliers are stable when refining (n_m, n_k, n_s) and tightening the DCT reduction threshold.

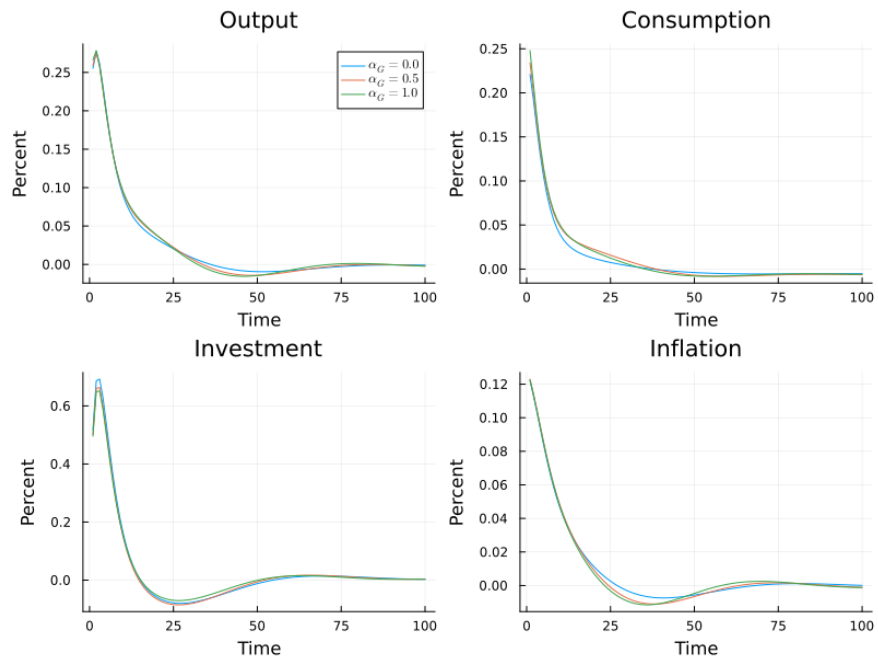
Implementation note. The solution is written in `Julia` and uses modular routines for Markov-chain construction, EGM policy updates for adjusters and non-adjusters, forward iteration of the distribution, DCT-based reduction and reconstruction, and Schur/QZ solution of the linearized system.

D Robustness Checks

D.1 Robustness to Government Spending in Utility

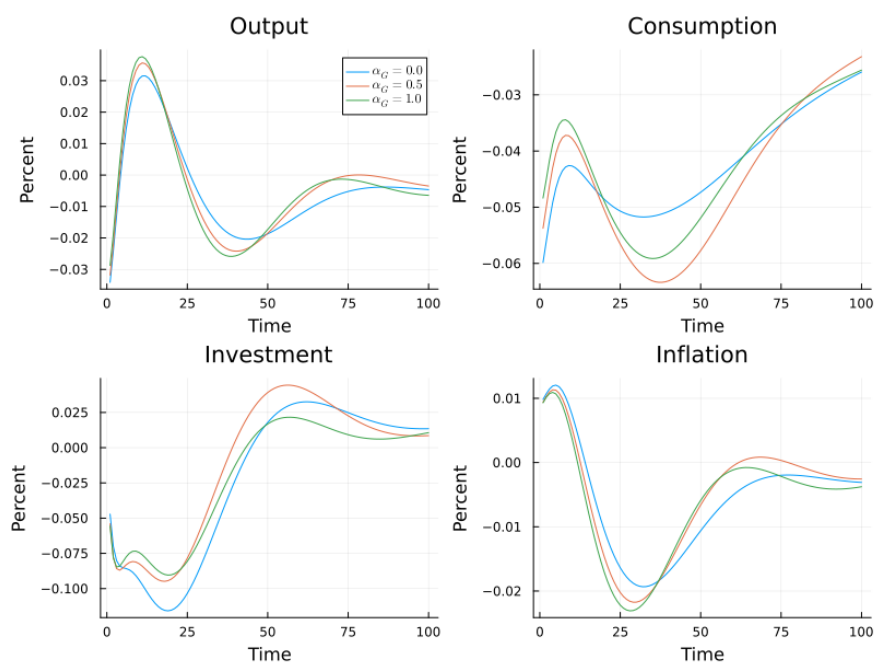
This appendix studies robustness to allowing government spending to enter household utility directly through $c + \alpha_G G$. I consider three values of α_G : 0, 0.5, and 1.0. Across all four fiscal shocks, the responses of output, consumption, investment, and inflation remain quantitatively very similar. The main results are therefore robust to this modification of preferences.

Figure 6: Robustness to government spending in utility: unfunded government purchases



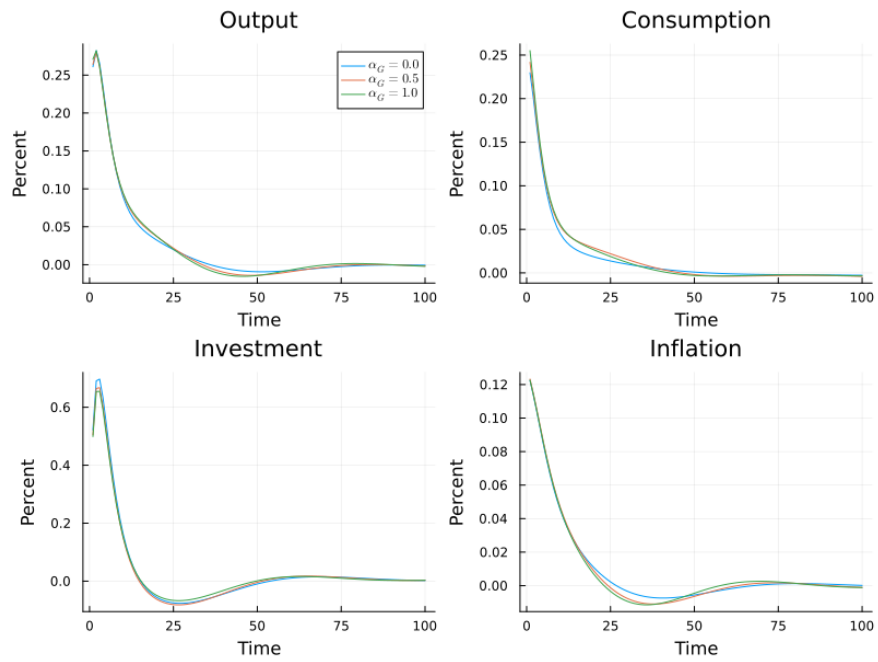
Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following an unfunded government-purchases shock, for alternative values of α_G .

Figure 7: Robustness to government spending in utility: funded government purchases



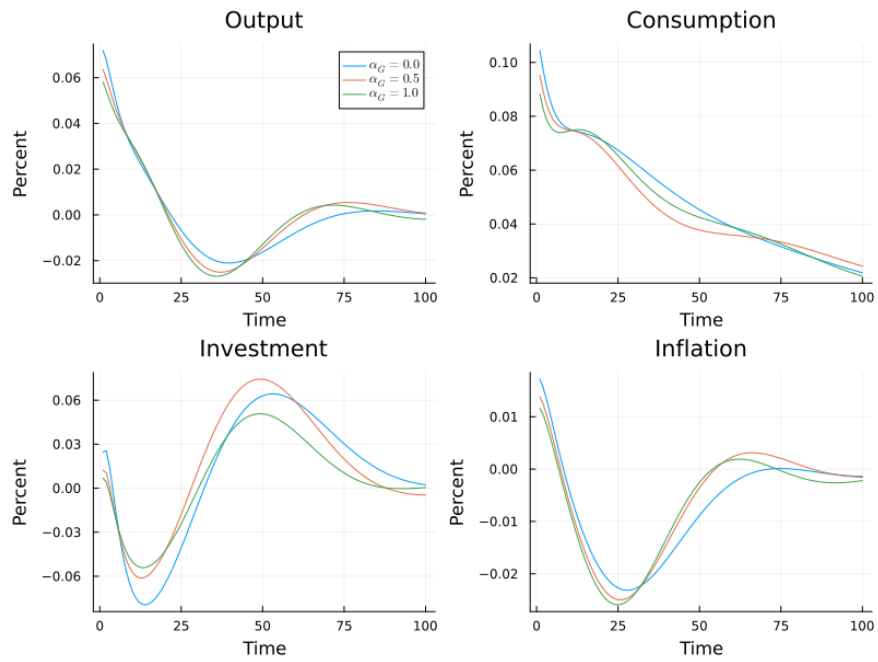
Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following a funded government-purchases shock, for alternative values of α_G .

Figure 8: Robustness to government spending in utility: unfunded transfers



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following an unfunded transfer shock, for alternative values of α_G .

Figure 9: Robustness to government spending in utility: funded transfers



Note: Impulse Response Functions (IRFs) for output, consumption, investment, and inflation following a funded transfer shock, for alternative values of α_G .